

A Level • AQA • Physics

 2 hours  14 questions

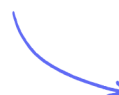
Structured Questions

Moments

Moments / Couples / Centre of Mass

Easy (5 questions)	/46
Medium (4 questions)	/40
Hard (5 questions)	/59
Total Marks	/145

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Easy Questions

1 (a) Write in words the equation to calculate the moment of a force.

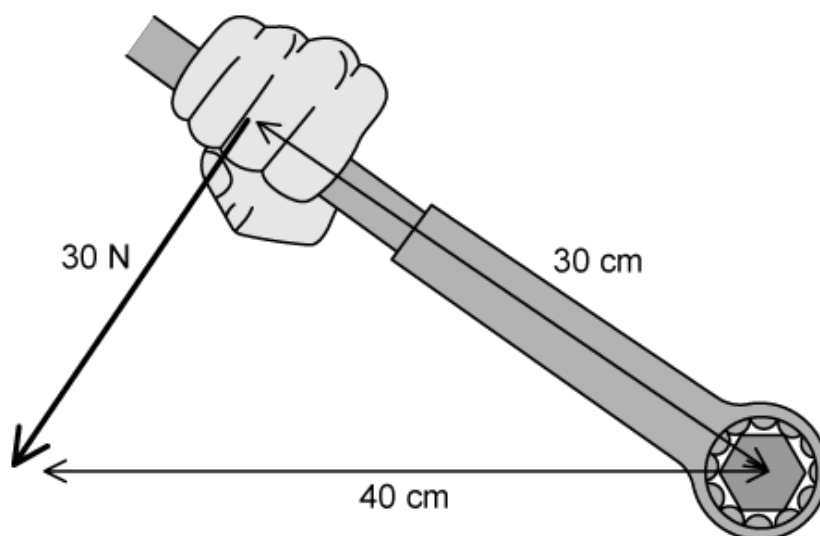
(1 mark)

(b) State whether a moment is a scalar or vector quantity and state the SI unit.

(2 marks)

(c) A force of 30 N is applied to the handle of a wrench tool as shown in **Figure 1**. The centre of the nut being loosened acts as a pivot.

Figure 1



Calculate the moment produced by the 30 N force.

(2 marks)

- (d) Suggest two ways the wrench operator in **Figure 1** could increase the moment produced on the wrench.

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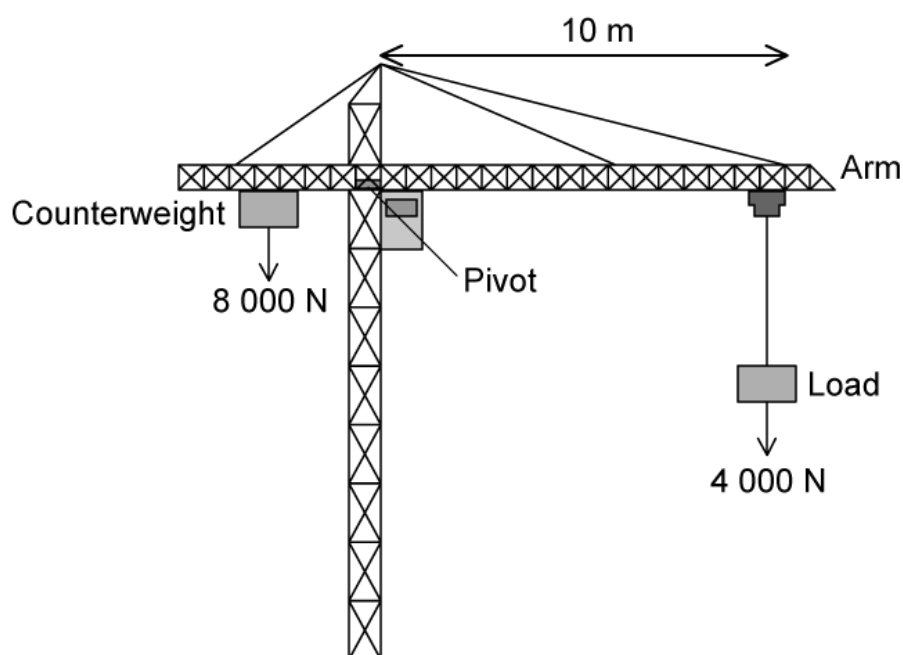
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(2 marks)

2 (a) State the principle of moments.

(3 marks)

(b) **Figure 1** below shows a crane lifting a load of 4 000 N. A counterweight of 8 000 N is used to keep the arm of the crane balanced when it is lifting the load.



Calculate the moment produced by the load and state whether this will act in a clockwise or anticlockwise direction.

(3 marks)

- (c) Calculate the distance of the counterweight force from the pivot.

Remember that the arm of the crane is balanced when it is lifting the load.

(3 marks)

- (d) The counterweight can be moved along the length of the crane arm. The position of the load and the size of the counterweight cannot be changed.

To enable the crane to lift heavier loads state the direction in which the counterweight should be moved and explain your answer.

(4 marks)

3 (a) State what is meant by the centre of mass of an object.

.....
(1 mark)

(b) The shapes shown in **Figure 1** below all have uniform density.

Figure 1



Draw an 'X' to indicate the position of the centre of mass of each shape.

.....
.....
.....
(3 marks)

(c) State **two** factors which affect the stability of an object.

.....
.....
(2 marks)

(d) The centre of mass and the centre of gravity do not always act through the same point.
Figure 2 shows the Earth and the moon.

Figure 2



Mark clearly the position of the centre of mass and centre of gravity **on the moon**. Label the centre of mass **M** and the centre of gravity **G**.

(2 marks)

- 4 (a)** The moment of a force about a point is defined as the magnitude of the force multiplied by the **perpendicular distance** from the line of action to the force.

Figure 1 shows a force, **F**, producing a moment about a point, **P**.

Figure 1

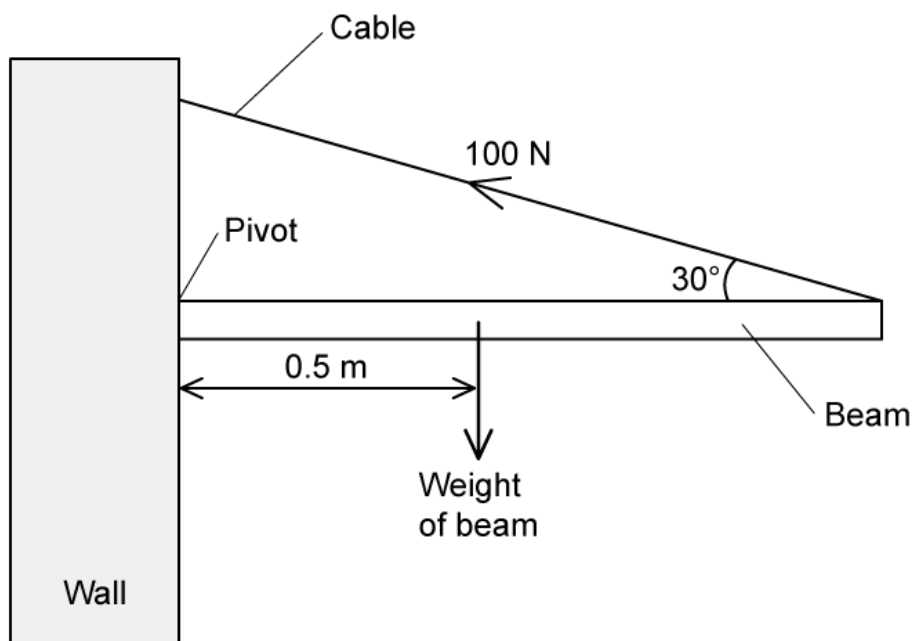


Show clearly on **Figure 1** what is meant by the perpendicular distance from the line of action of the force to the point **P**. Label the perpendicular distance **d**.

(1 mark)

- (b)** **Figure 2** below shows a beam of length 1 m, which is held in equilibrium against a wall by a cable which provides a tension force of 100 N. The cable makes an angle of 30° to the beam.

Figure 2



By **resolving** the 100 N force show that the vertical component of the force provided by the cable is 50 N.

(2 marks)

- (c) Using your answer to part (b) calculate the moment provided by the cable and state whether the direction of this moment is clockwise or anticlockwise.

(3 marks)

- (d) Use the principle of moments to calculate the weight of the beam.

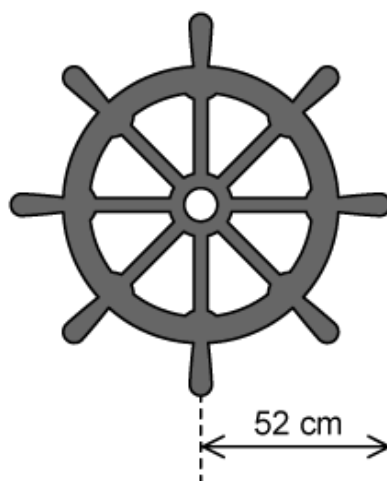
(3 marks)

- 5 (a) State **three** conditions which must apply to a pair of coplanar forces to make them a couple.

(3 marks)

- (b) A ship has a steering wheel of radius 52 cm, as shown in **Figure 1**. Under rough sea conditions the moment of the couple needed to turn the wheel is 38 N m.

Figure 1



Assuming that the crewman steering the ship places both hands on the steering wheel in order to produce a couple, draw on **Figure 1** the positions and directions of the forces needed to turn the wheel **anticlockwise**.

(2 marks)

- (c) Calculate the magnitude of the force exerted by each hand of the crewman in order to turn the ship's wheel under rough sea conditions.

(2 marks)

- (d)** The crewman takes one hand off the steering wheel and continues to apply the same force with the other.

Use your answer to part (c) to calculate the moment produced by the hand which remains on the steering wheel.

(2 marks)

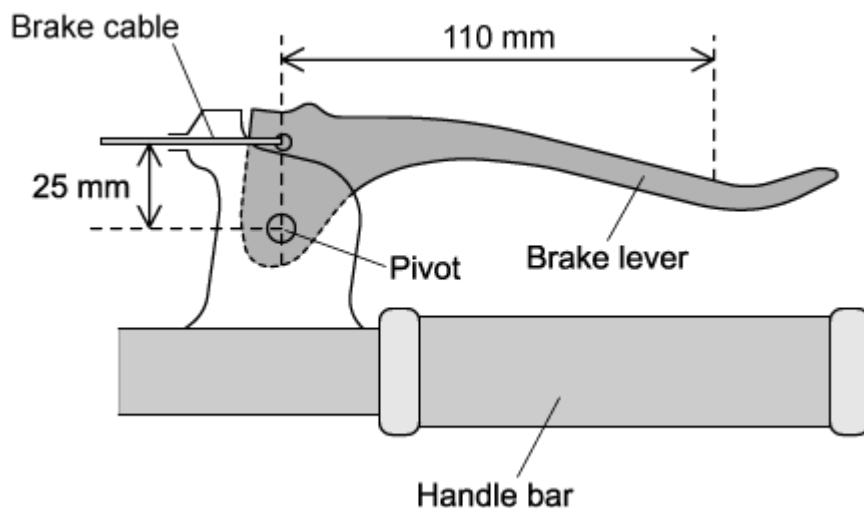
Medium Questions

1 (a) State the principle of moments.

(3 marks)

(b) **Figure 1** shows a bicycle brake lever that has been pulled with a 50 N force to apply the brake.

Figure 1



Draw two arrows on **Figure 1** to show the directions of the tension in the brake cable and the force applied by the cyclist on the brake lever.

(2 marks)

- (c) Calculate the moment of the force applied by the cyclist about the pivot. State an appropriate unit.

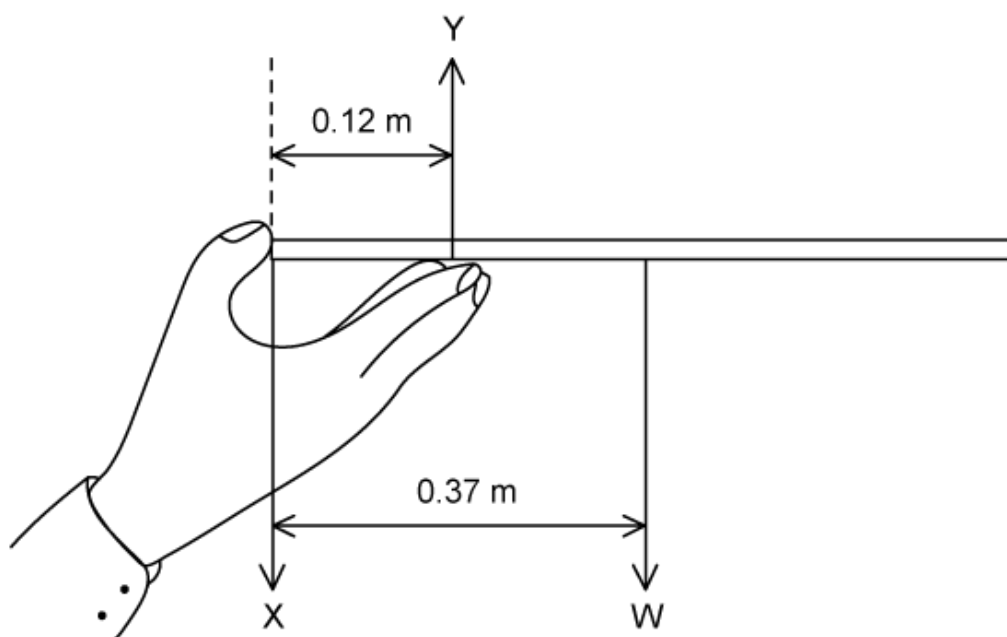
(3 marks)

- (d) Calculate the tension in the brake cable. Assume the weight of the lever is negligible.

(3 marks)

- 2 (a)** A waiter holds a uniform tray of mass 150 g horizontally in one hand between their fingers and thumb as shown in **Figure 1**.

Figure 1



X , Y and W are the three forces acting on the tray.

Describe the relationships between the forces that must be satisfied if the tray is to remain horizontal and in equilibrium.

(2 marks)

- (b)** Calculate the length of the tray. Explain any assumptions made.

(3 marks)

(c) Calculate the magnitude of the force X and Y .

(4 marks)

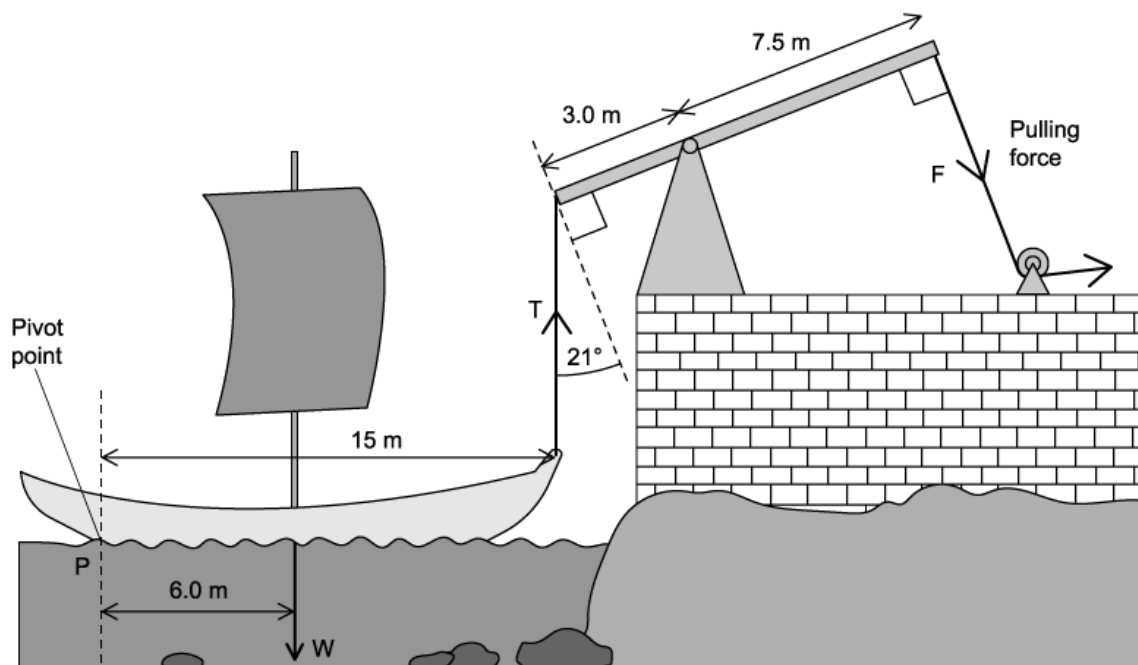
(d) The waiter places a glass of water on the tray.

Explain where the glass should be position on the tray if the force X is to have the same value as in part (c).

(2 marks)

- 3 (a)** It is said that Archimedes used huge levers to sink Roman ships invading the city of Syracuse. A possible system is shown in the **Figure 1** where a rope is hooked on to the front of the ship and the lever is pulled by several men.

Figure 1



Calculate the mass of the ship if its weight is $4.6 \times 10^4 \text{ N}$.

(1 mark)

- (b)** Calculate the moment of the ship's weight about point **P**. State an appropriate unit for your answer.

(2 marks)

- (c)** Calculate the minimum vertical force, T , required to start to raise the front of the ship. Assume the ship pivots about point **P**.

(2 marks)

- (d) Calculate the minimum force, F , that must be exerted to start to raise the front of the ship.

(3 marks)

4 (a) The torque of a couple is given by

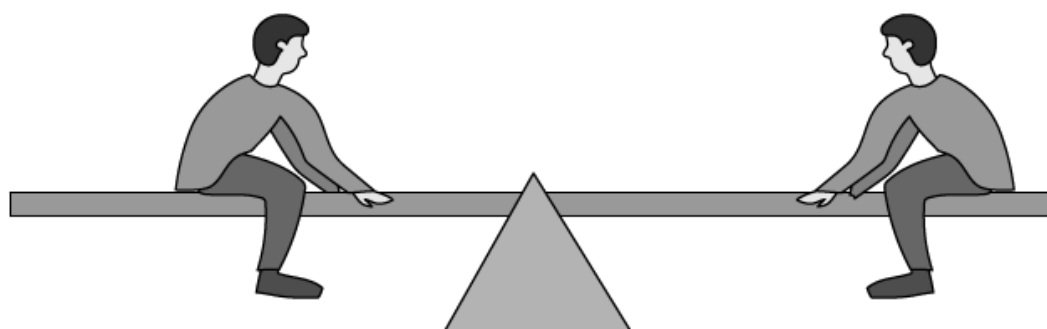
$$\text{torque} = F \times s$$

With the aid of a diagram explain what is meant by a couple. Label F and s on your diagram. State the appropriate unit for the torque of a couple.

(4 marks)

(b) The see-saw shown in **Figure 1** consists of a uniform beam freely pivoted at the centre of the beam. Two children sit opposite each other so that the see-saw is in equilibrium.

Figure 1



Explain why the see-saw is in equilibrium.

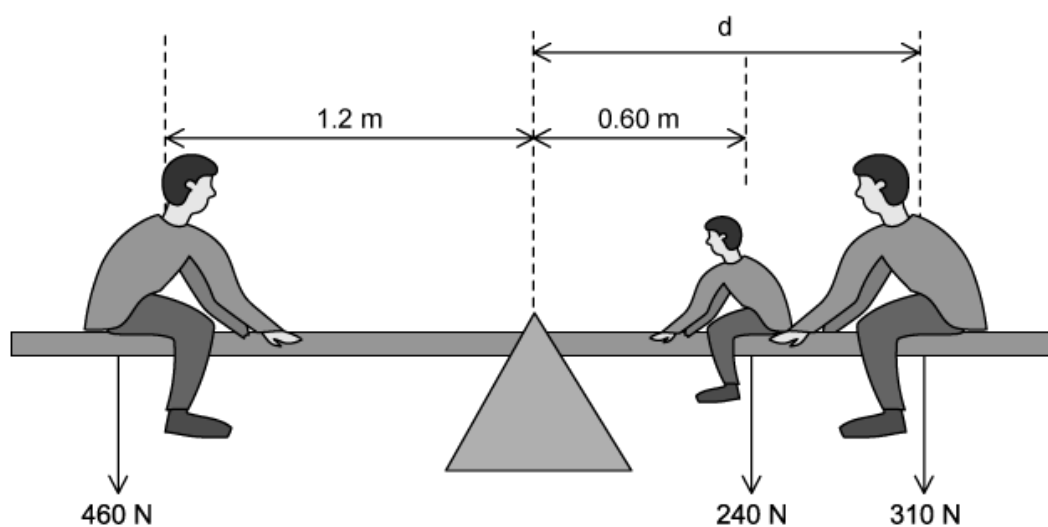
(2 marks)

(c) Explain why the weight of the beam does not affect equilibrium.

(2 marks)

- (d) **Figure 2** shows the see-saw with three children of weights 460 N, 310 N and 240 N sitting so that the see-saw is in equilibrium.

Figure 2

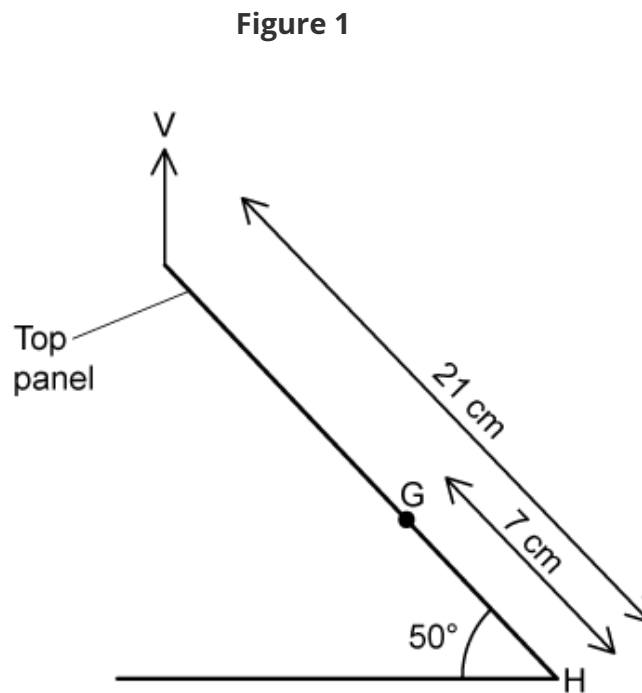


Calculate the distance, d .

(2 marks)

Hard Questions

- 1 (a) **Figure 1** shows the opening of the top panel of a laptop. The panel is held open at an angle of 50° to the horizontal by a vertical force V applied at one end of the panel. The panel is 21 cm long, has a mass of 0.5 kg and its centre of gravity, G is 7 cm from the hinge at H .



Calculate the magnitude of the force acting at the hinge H .

(3 marks)

- (b) At an angle of 65° , in order to open the top panel of the laptop further, V must now act perpendicular to the top of the panel.

Calculate the magnitude of the new force acting at the hinge H .

(4 marks)

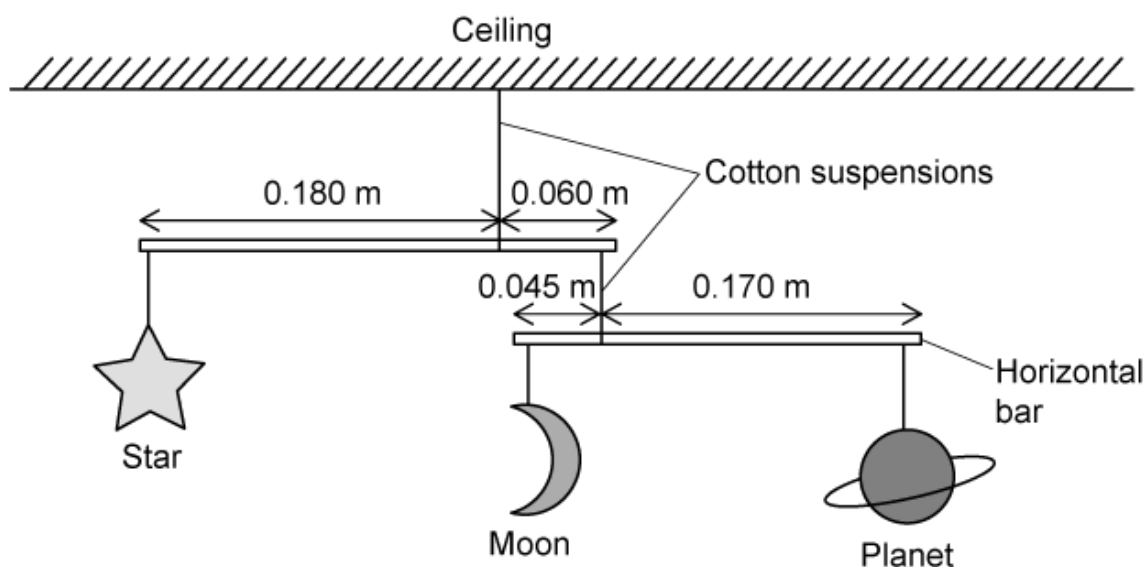
- (c)** The largest angle the top panel of this laptop can open to is 90° .

Determine whether it would be easier for someone to continuously apply the force on the top panel, V vertically upwards on the panel, like in part (a), or continuously perpendicular to the panel, like in part (b).

(3 marks)

- 2 (a)** Figure 1 shows a simple toy mobile. This is a suspended ornament that rotates and swings in a slight breeze. A star, moon and planet toy are suspended by cotton suspensions from the ceiling.

Figure 1



The weights of the objects are such that they are in equilibrium, with the weight of the planet as 1.35 N. The weights of the horizontal bars are negligible.

Calculate the weight of the star.

(4 marks)

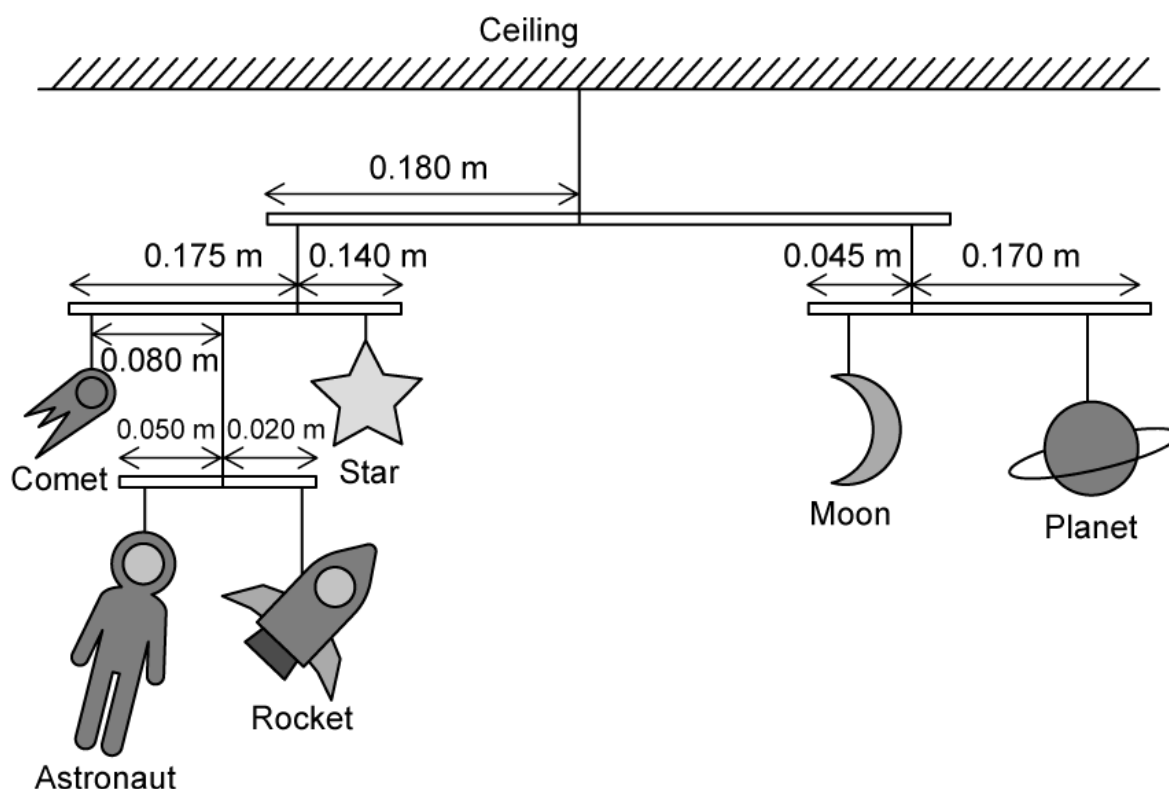
- (b)** A child pulls vertically downward on the moon toy.

Explain the effect this would have on the equilibrium of the system.

(3 marks)

- (c) A different mobile is set up in **Figure 2** which is also in equilibrium. Added objects are a comet, astronaut and rocket.

Figure 2



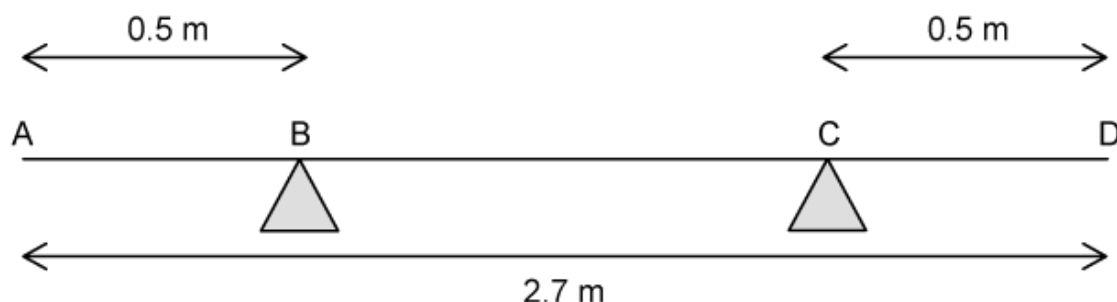
The weight of the rocket is 1.50 N. The weights of the horizontal bars are still negligible.

Calculate the distance between the suspension hanging the moon and planet to the suspension hanging the first bar from the ceiling.

(4 marks)

- 3 (a)** A bench consists of a plank which is resting in a horizontal position on two supports **B** and **C**. The plank is modelled as a uniform rod **AD** of length 2.7 m and mass 15 kg. The supports at **B** and **C** are 0.5 m from each end of the plank, as shown in **Figure 1**.

Figure 1



Two students, Rose and Quinn, sit on the plank. Rose has mass 55 kg and sits in the middle of the plank and Quinn has a mass 62 kg and sits at the end **A**. The plank remains horizontal and in equilibrium. The students can be modelled as point masses.

Calculate the magnitude of the normal reaction forces at **B** and **C**.

(4 marks)

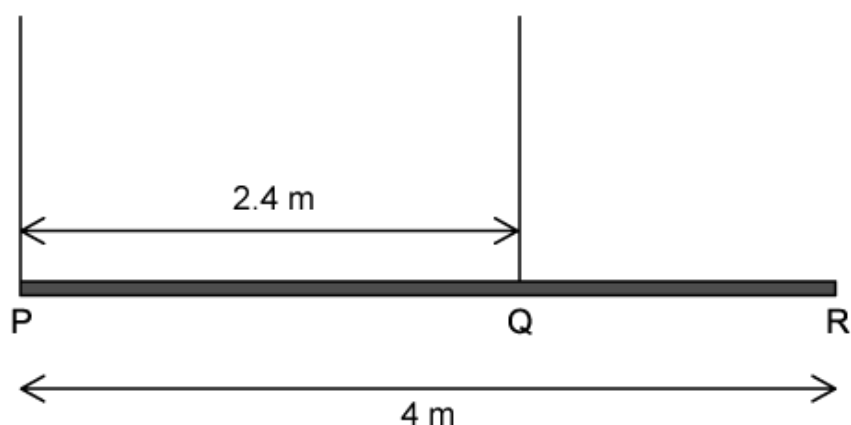
- (b)** Whilst Quinn stays at point **A**, Rose now moves along the plank to a point **X**.

Given that the plank remains horizontal and in equilibrium, and that the magnitude of the normal reaction force at **B** is now twice the magnitude of the normal reaction force at **C**, calculate the distance **BX**.

(4 marks)

- (c) A pole **PR** has a length 4 m and a weight W newton. The pole is held in a horizontal position in equilibrium by two vertical ropes attached to the pole at the points **P** and **R** where **PR** = 4 m, as shown in **Figure 2**.

Figure 2



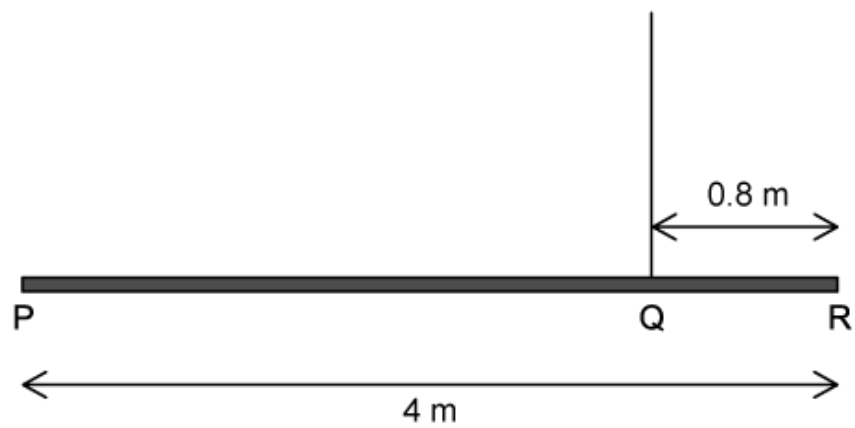
A load of weight 12 N is attached to the rod at **R**. The pole is uniform and the ropes are light inextensible strings.

Given that the tension in the rope attached to the pole at **Q** is seven times the tension in the rope attached to the pole at **P**, calculate the value of W .

(4 marks)

- (d) **Q** is now moved to a point further along the pole until the length **QR** is 0.8 m, as shown in **Figure 3**. The pole still remains in equilibrium. The load of 12 N is now moved to a point x metres from **P**. The beam remains in equilibrium in a horizontal position.

Figure 3



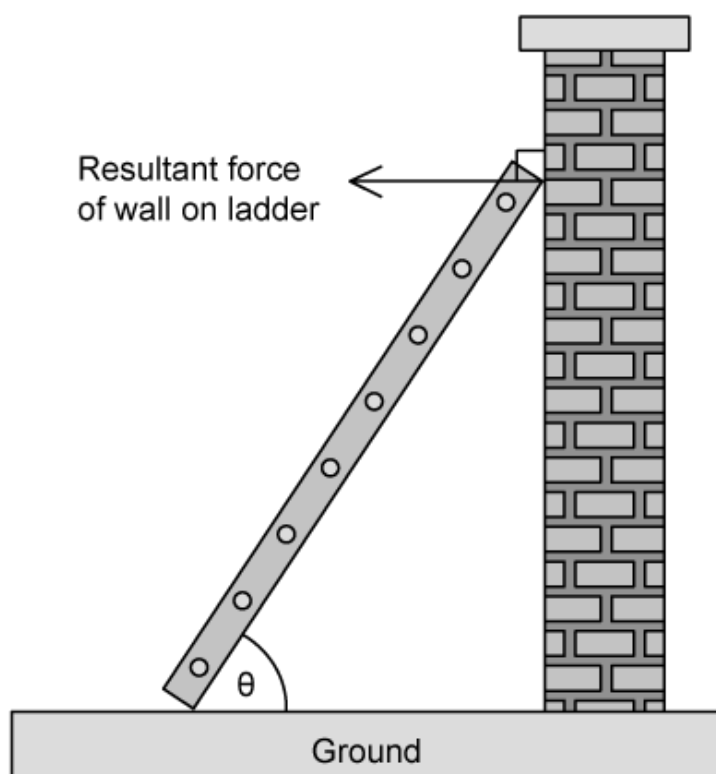
The rope at **Q** will break if its tension exceeds 154 N. The rope at **P** cannot break.

Calculate the range of possible positions on the beam where the load can be attached without the rope at **Q** breaking.

(3 marks)

- 4 (a) **Figure 1** shows a uniform ladder resting against a vertical wall. The ladder is at an angle of θ to the ground.

Figure 1



The resultant force from the wall on the ladder is twice the weight of the ladder.

Show that the angle θ is around 14° .

You may use the trigonometric identity: $\tan \theta = \frac{\sin \theta}{\cos \theta}$.

(3 marks)

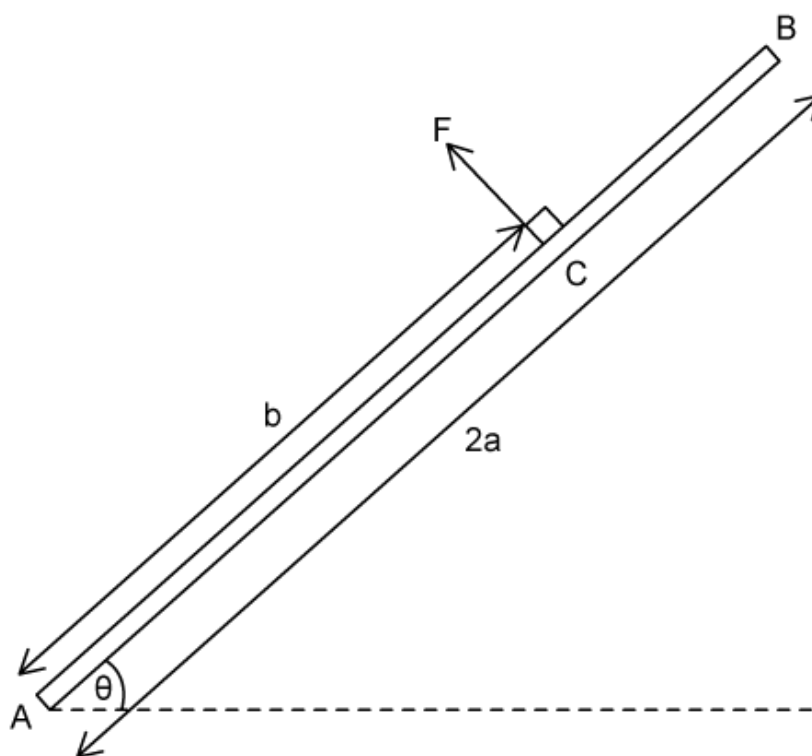
- (b) Suggest the changes to the forces acting on the ladder that occur when someone climbs it.

(3 marks)

- (c) A uniform rod **AB** of mass m and length $2a$ is freely hinged to a fixed point **A**. An object of mass m is attached at **B** and the rod is held in equilibrium at an angle θ to the horizontal by a force of magnitude F acting at point **C** on the rod.

The force acts at right angles to **AB** where **AC** = b as shown in **Figure 2**.

Figure 2



In terms of a , b , m and θ , determine the equations for the horizontal and vertical components of the force acting on the rod at **A**.

(3 marks)

- (d) Given that mass m is 2 kg, a is 4 m and b is 1.3 m and the angle θ is 70° , show that the resultant force acting on the rod at **A** is around 61 N.

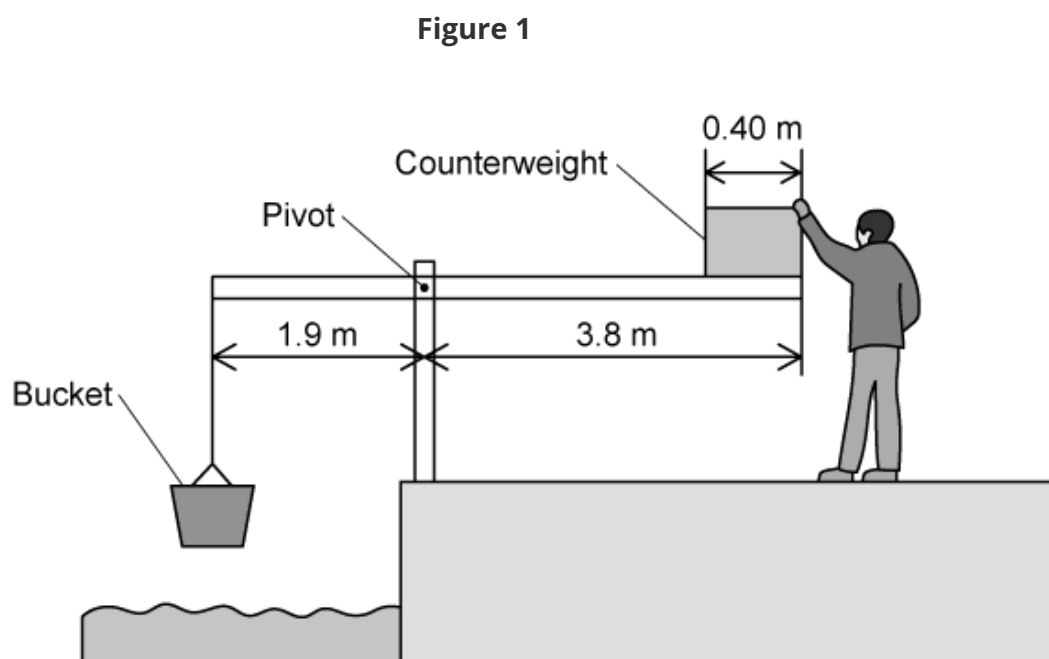
(3 marks)

- 5 (a) A metre ruler has a support pivot 0.20 m from one end. The centre of mass of the ruler lies 0.43 m from this end. A 1.5 kg mass is positioned 0.30 m from the opposite end of the ruler to the pivot. An upwards force of 30 N is applied at this end of the ruler at an angle of 20° .

Calculate the mass of the ruler.

(3 marks)

- (b) A shaduf is a device used to lift water from a well. It consists of an upright support to which a uniform beam is pivoted. It can be assumed that the weight of the beam is negligible. On one end of the beam is a counterweight, and on the other a bucket which can hold water, as shown in **Figure 1**.



The counterweight is of uniform material and has a mass of 7.1 kg and is 0.40 m long. The bucket has a weight of 165 N and a capacity of 0.26 m^3 . When the bucket is half full, a force is required at the end of the beam to lift the bucket and water.

Calculate the value of this force when the beam is horizontal.

Density of water = 1 g cm^{-3}

(4 marks)

- (c) A student says that if the weight of the beam is **not** considered to be negligible, then the force at the end of the beam would become larger to counteract this weight and keep the beam in equilibrium.

Discuss whether the student is correct.

(4 marks)